

Skills Summary

Finding Parameters That Make Piecewise Functions Continuous

Use this reference while completing your worksheet or when studying.

The continuity test at $x = p$. f is continuous at p when (i) $f(p)$ is defined, (ii) $\lim_{x \rightarrow p} f(x)$ exists, and (iii) the two are equal. On a piecewise function only the *junctions* can fail — each piece is already continuous on its own interval.

Turn it into an equation: left-hand limit = right-hand limit = $f(p)$. One junction gives one equation, so one junction determines one unknown constant and two junctions determine two.

1. Solve the junction equation for a constant

Substitute the junction value p into the formula on each side — for polynomial and rational pieces the one-sided limit is just that substitution. Set the two results equal and solve. With two unknowns, start at the junction that involves only one of them.

Example: $f(x) = 3x + a$ for $x < 4$ and $f(x) = 2x + 7$ for $x \geq 4$. Find a .

Step 1. Both pieces are polynomials, so the only risk is the junction at $x = 4$ — set the one-sided limits equal there.

Step 2. Left: $3(4) + a = 12 + a$. Right: $2(4) + 7 = 15$, which is also $f(4)$. So $12 + a = 15$.
 $\Rightarrow \mathbf{a = 3}$

Try it: $f(x) = x^2 + a$ for $x < 3$ and $f(x) = 4x - 1$ for $x \geq 3$. Find a .

check: $a = 8$

2. Fill a removable hole with the limit value

When a piece is defined for $x \neq p$ and a constant k is assigned at p , condition (iii) reads $k = \lim_{x \rightarrow p} f(x)$ — so the whole problem is a limit problem.

Substitution giving $\frac{0}{0}$ means the form is indeterminate, not that the limit is 0: factor and cancel, multiply by the conjugate for radicals, or use a known limit such as $\lim_{u \rightarrow 0} \frac{e^u - 1}{u} = 1$.

Example: $f(x) = \frac{x^2 - 25}{x - 5}$ for $x \neq 5$, and $f(5) = k$. Find k .

Step 1. Continuity at a hole forces k to be the limit, and substitution gives $\frac{0}{0}$ — so factor and cancel first.

Step 2. $\frac{(x - 5)(x + 5)}{x - 5} = x + 5$ for $x \neq 5$, and substituting $x = 5$ into $x + 5$ gives 10.
 $\Rightarrow \mathbf{k = 10}$

Try it: $f(x) = \frac{x^2 - 16}{x - 4}$ for $x \neq 4$, and $f(4) = k$. Find k .

check: $k = 8$

3. Show that no constant can work

A constant supplies $f(p)$ only. If the *limit* fails to exist, no choice of k can repair the function. Compute both one-sided limits. Two different finite values means a jump discontinuity; an infinite one-sided limit means an infinite discontinuity. Both are non-removable, and the honest answer is “no such k exists”.

Example: $f(x) = x + 1$ for $x < 0$ and $f(x) = x^2 - 3$ for $x > 0$, with $f(0) = k$. Can any k make f continuous at $x = 0$?

Step 1. The value k only settles condition (i), so test condition (ii) first: does the limit exist?

Step 2. From the left, $x + 1 \rightarrow 1$. From the right, $x^2 - 3 \rightarrow -3$.

The one-sided limits are **1** and **-3**; they differ, so the limit does not exist and no k works — a jump discontinuity.

Try it: $f(x) = 2x - 5$ for $x < 1$ and $f(x) = x + 1$ for $x > 1$. Find both one-sided limits at $x = 1$ and decide whether some k works.

check: **2** and **3** — they differ, so no k works